Simulating Exposure for xVA

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Assignment presented in the partial fulfilment

of the requirement for the course

xVA

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CHAPTER 1

INTRODUCTION

Modelling the future exposure of a trade is important in derivatives valuation and pricing, as it is key in modelling counterparty credit risk. The most robust methodology for this is Monte Carlo simulation due to its flexibility and accuracy. However, as the number of risk factors, assets, and counterparties increases, Monte Carlo simulation becomes highly computationally intensive. This is a problem for traders who need accurate and profitable quotes in real time. Consequently, this project explores methods that accelerate the future exposure simulation.

CHAPTER 2

EXPOSURE

Parties that trade derivatives with other counterparties through bilateral agreements are exposed to the risk that their counterparty defaults, known as counterparty credit risk (CCR). The quantification of CCR is done through a credit valuation adjustment (CVA), one of the xVA's. As we will see, a party's exposure to its counterparty plays a key role in this quantification (Gregory, 2020).

2.1 DEFINING EXPOSURE

According to Green (2015), exposure at default (EAD) is the amount owed by the defaulting party at the time of default. This is analogous to positive exposure, which is defined mathematically in Gregory (2020) as:

$$\text{Positive Exposure} = \max(V, 0),$$

where V is the value of the derivative at the time of default. There is a clear asymmetry here, since the party is only exposed when they are owed money. If the derivative is a liability, they have no exposure. In the case of a party owing money when the counterparty defaults, the party is still required to pay. Hence, it is also useful to look at the negative exposure:

$$\text{Negative Exposure} = \min(V, 0).$$

2.2 CALCULATING EXPOSURE

If the contract is collateralised, then the equation becomes:

$$\text{Positive Exposure} = \max(V - M_{received} + M_{posted}, 0).$$

If there are multiple transactions with the same counterparty, netting is also applied to obtain the total value. Calculating the current exposure is done by substituting the mark-to-market value of the derivative (or netted value of derivatives) into the equation and accounting for margin. The current exposure is easy to calculate, as all variables are known. In the context of CVA, however, we want to model what exposure looks like in the future. This is much harder, as it requires probabilistic modelling of risk factors, margining and netting. The difficulty is illustrated in the following graph:

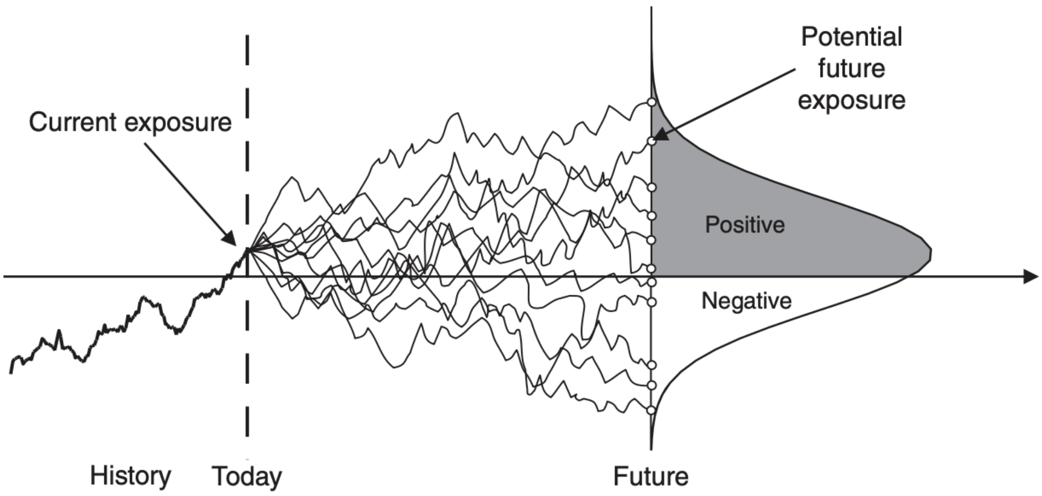

Figure 2.1: Modelling future exposure (Gregory, 2020)

From this, we introduce the expected positive exposure as in Green (2015):

$$EPE(t) = \mathrm{E}\left[\max(V, 0) \mid \mathcal{F}_t\right]$$

This is the expected positive exposure at some future date t given all information available at t, i.e. the average of all possible positive exposure values at time t. It is important to note that the zeros are included in the average calculation here. EPE is a function of t, thus calculating it over time gives us an exposure profile for the derivative or netting set. This is useful for CCR modelling purposes. For example, the formula for approximating unilateral CVA as given in Gregory (2020) is:

$$UCVA(t) \approx -LGD \sum_{i=1}^{m} EPE\left(t, t_i\right) \times PD\left(t_{i-1}, t_i\right),$$

where LGD is the percentage loss given the counterparty defaults, $EPE\left(t, t_i\right)$ here is the discounted expected positive exposure at time point t_i (usually discounted at the risk-free rate), and $PD(t_{i-1}, t_i)$ is the probability of the counterparty defaulting between point t_{i-1} and point t_i. It is important to note that the derivative value V is itself usually an expected value and a function of time, i.e.:

$$V(t) = \mathrm{E}_t[g(X)],$$

where $g(X)$ is a discounted payoff (or cash flow) function. For example, a European call option has the function $g(X) = \max(S_T - K, 0)\exp(-\int_t^T r(s)ds)$. Depending on the derivative (or the model used to price it), this expectation may not have a closed-form solution.

Due to the probabilistic nature of calculating expected exposure, the most robust way of calculating the exposure profile is using Monte Carlo simulation. It applies to many different scenarios with different netting sets, risk factors and dimensions. The methodology for this is explained in the next chapter.

2.3 OTHER EXPOSURE METRICS

In addition to EPE, there are two other important exposure metrics. The first is expected negative exposure (ENE):

$$ENE(t) = \mathrm{E}\left[\min(V, 0) \mid \mathcal{F}_t\right].$$

This is equivalent to the counterparties EPE calculation on the party (Green, 2015). This is used in the debit value adjustment calculation (DVA), which captures the parties own benefit of defaulting. The second metric is the expected future value (EFV):

$$EFV(t) = \mathrm{E}\left[V \mid \mathcal{F}_t\right].$$

This is just the expected value of the derivative at time t. It is used in the funding value adjustment calculation (FVA). The methods explored in the next sections are not just useful for EPE; they can easily be used to calculate ENE and EFV. Further, the general formula for xVA as given in Gregory (2020) is:

$$xVA = \int_0^\infty C(t)D(t)E[X(t)]dt$$

Our focus is on $E[X(t)]$, the expected utilisation component. In CVA, this is the EPE. This term generally needs to be modelled, so this methodology can be applied to all xVA term calculations, not just CVA. The utilisation terms for each xVA calculation is summarised in the table below:

Table 2.1: xVA terms and their utilisation components

Term	**Area**	**Utilisation component**	**Calculation**
CVA	Credit	EPE	$\max(\text{value}, 0)$
DVA	Credit	ENE	$\min(\text{value}, 0)$
FVA	Funding (and margin)	EFV	value
FCA	Funding (and margin)	EPE	$\max(\text{value}, 0)$
FBA	Funding (and margin)	ENE	$\min(\text{value}, 0)$
KVA	Capital	ECP	Function of value
MVA	Initial margin	EIM	

CHAPTER 3

SIMULATING EXPOSURE

3.1 MONTE CARLO RECAP

The following explanation is based on Rizzo (2019). Say we want to estimate θ:

$$\begin{aligned} \theta &= \int_{-\infty}^{\infty} f(x)g(x)dx \\ &= \mathrm{E}_X[g(X)] \end{aligned}$$

where $f(x)$ is the probability density function of the random variable X. Then the Monte Carlo estimate is:

$$\hat{\theta} = \frac{1}{N}\sum_{n=1}^{n} g(x_n),$$

where $x_1, x_2, ..., x_N$ are independent and identically distributed simulated observations of X.

3.2 MONTE CARLO FOR EXPECTED POSITIVE EXPOSURE

Monte Carlo is used for calculating EPE, as it is flexible enough to model portfolio effects, path dependence, collateralisation, trade-specific details and high dimensionality. Other methods, such as parametric and semi-analytical, are available; however, Monte Carlo is considered "state-of-the-art" (Gregory, 2020). Based on the previous section, we can write the Monte Carlo estimate of EPE as follows:

$$E\hat{P}E(t) = \frac{1}{N}\sum_{n=1}^{N} \max(\hat{V}^{(n)}(t,T), 0),$$

where $\hat{V}^{(n)}(t,T)$ is the estimated price from simulation n at time t of the derivative with maturity T. Here, the Monte Carlo problem is now multi-step. There are various models to choose to simulate the random variables, based on portfolio complexity, dimension and computational expense. The derivative price is also based on an underlying model. For simplicity purposes we stick to modelling equity prices with a geometric Brownian motion (GBM) and interest rate with the Hull-White one-factor model (HW1F). To get an exposure profile over time, we require a multi-step Monte Carlo simulation (Green, 2015). We define the notation:

- Time steps: $t \in \{1, 2, ..., T\}$
- Simulation Paths: $n \in \{1, 2, ..., N\}$

The procedure is as follows:

1. Simulate states of the world
    - N paths
    - Each path has T time steps
    - This results in a N by T matrix
2. Price derivative at each state
    - This results in a N by T matrix of prices
3. Calculate $E\hat{P}E(t)$ $\forall t \in \{1, 2, ..., T\}$
    - Set the negative prices in the price vector to 0
    - Take the average of the columns.
    - This gives a T-length vector of EPE values that can be used to calculate CVA

To do this at the netting set level, instead of the trade level, we use:

$$\mathrm{V}_{\mathrm{Agg}}^{(t)}(t) = \sum_{k=1}^{K} V^{(n)}(k, t),$$

where $k \in \{1, 2, ..., K\}$, with K being the number of trades in the netting set.

3.3 THE COMPLEXITIES OF SIMULATING EPE

The issue with Monte Carlo is that it can be computationally expensive, particularly in xVA. The reasons for such are talked about further.

3.3.1 Dimensionality

Using an example from Gregory (2020), if there are 250 counterparties each with 40 derivatives and we simulate 10000 paths with 100 time steps, that is 10 billion trade revaluations. In general, the time complexity will be:

$$\mathcal{O}(TNKC)$$

where C is the number of counterparties and T, N and K are defined as before. This is less of an issue for CVA on the netting set level, however, the example still illustrates how Monte Carlo scales and the broader issue of Monte Carlo in xVA. Typically, 10 000 simulations are used and there needs to be enough time points to correctly capture the exposure profile (Gregory, 2020). Focusing on a single counterparty, there may be several risk factors that need to be modelled, not just one (e.g several FX and interest rate models).

3.3.2 Model Complexity

When it comes to model choice, there is a trade-off between accuracy and speed. More complex models are required for better accuracy. For example, modelling interest rates are often done using the HW1F models, as it is analytically tractable. However, it can't calibrate to all implied volatilities and allows negative rates. The choice of model must therefore balance simulation speed and accuracy. An additional issue that must be considered is calibration. More complex models are usually harder to calibrate. The complexity doesn't just apply to risk factors. For example, accounting for collateral in the EPE simulation requires additional calculations (Gregory, 2020).

3.3.3 Pathwise v.s. Direct

Gregory (2020) describes two methods of simulation in the context of xVA: pathwise and direct. The pathwise approach simulates the points of time sequentially, with each point being based on the previous. This means the simulations are dependent. This is a natural approach, as it gives an accurate description of possible paths. The direct method, on the other hand, simulates potential points of default first, and then the exposure is simulated at that point (i.e. the time grid is not fixed). This usually converges quicker than the pathwise approach. The direct approach can be used when only a single value is needed for the xVA, not a full profile. As a result, we focus on pathwise here. Additionally, any path dependency in the EPE calculation, like variation margin, requires the full path of movements and therefore the path-wise approach. The difference between the two methods is visualised in the following image:

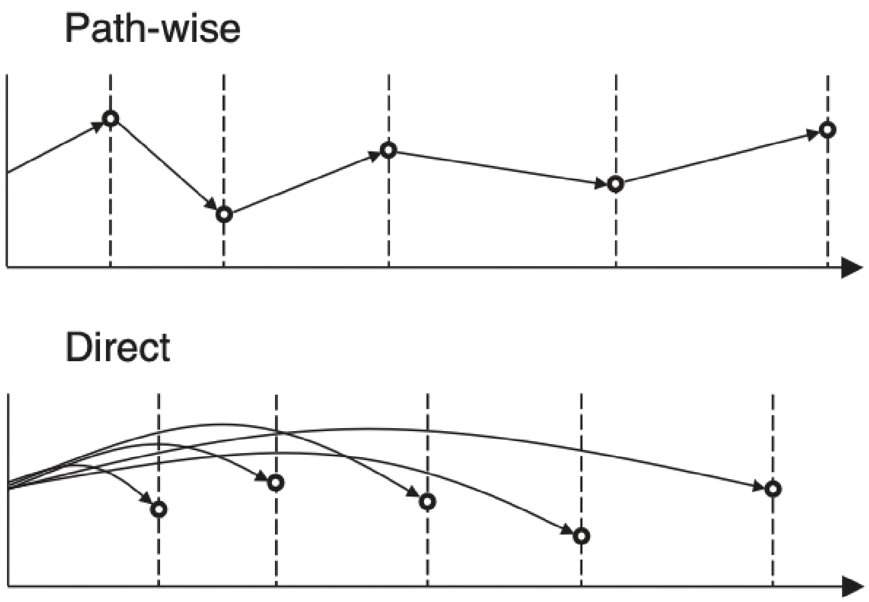

Figure 3.1: Path-wise v.s. Direct Simulation (Gregory, 2020)

3.3.4 Nested Monte Carlo

Nested Monte Carlo, as described in Green (2015), can be thought of as a simulation in a simulation. To explain how this arises, consider our Monte Carlo estimate from before:

$$\theta = \mathrm{E}_X[g(X)]$$

A nested simulation arises if $g(X)$ itself is an expected value that requires. This arises in an

EPE simulation if the derivative value $V(t, T)$ does not have a closed-form solution and itself must be estimated with a Monte Carlo simulation. This issue arises when the model chosen for the derivative doesn't have a closed-form solution or the derivative itself is complex or path-dependent. Examples of such derivatives are Asian, American and Bermudan options. This results in a modified simulation procedure:

1. Simulate states of the world
    - N_{outer} paths
    - Each path has T time steps
    - This results in a N by T matrix
2. Price derivative at each state
    - Each state requires a new simulation of N_{inner} paths
    - Each new path has $(T - t)$ time steps, depending on the time step we are at
    - Calculate price at state base on inner simulation
3. Calculate $E\hat{P}E(t)\ \forall t \in \{1, 2, ..., T\}$
    - Set the negative prices in the price vector to 0
    - Take the average of the columns.
    - This gives a T-length vector of EPE values that can be use to calculate CVA

The Monte Carlo procedure from before had a time complexity of $\mathcal{O}(TN)$. In the nested case, each price calculation has its own time complexity of $\mathcal{O}((T - t)N_{inner})$. The resulting simulation is thus extremely computationally expensive. This doesn't only apply to EPE but to any xVA simulation where the estimate is dependent on future paths. For example, expected initial margin in the MVA calculation requires a simulation at each point to calculate a value at risk-type measure – this is a nested simulation. This also applies if we model initial margin through time in our EPE simulation.

CHAPTER 4

IMPROVING SIMULATION EFFICIENCY

Monte Carlo Simulation has a rich literature of methods for improving the simulation. They can be viewed from a statistical perspective, with the aim of getting a more accurate result in the same amount of simulations, or the computational perspective, with the aim of achieving a given accuracy with the least amount of simulations. These are two sides of the same coin, and in xVA we are concerned with both accuracy and speed. In the next sections we explore some methods of improving the EPE simulation.

4.1 ANTITHETIC VARIABLES

The explanation of antithetic variables is based on Rizzo (2019). The variance of estimates using Monte Carlo decreases as the number of simulations increases. In other words, increasing the number of simulations results in more accurate answers. This is not always practical, as made clear by the dimensionality issues explained in section 3.3.1. The goal of variance reduction is essentially to get accurate estimates in fewer simulations. Compared to a crude EPE simulation, this means we could get the same accuracy in less simulation or increase accuracy with the same amount of simulations, depending on what the focus is.

A popular method of variance reduction is antithetic variables. The main idea is to simulate a set of $\frac{N}{2}$ random variables and use them to get a second modified set of $\frac{N}{2}$ random variables. The modification step forces the second set to be negatively correlated to the first. For example, for a uniform random variable U_1, you can use $U_2 = 1 - U_1$ as that is still uniform and negatively correlated. The case of normal random variables is described in Zhang (2009). The crude estimate is the same as before:

$$\bar{\theta} = \frac{1}{n}\sum_{i=1}^{n} g\left(X_i\right)$$

$$\text{with i.i.d. } X_i \sim N(0,1),$$

The antithetic estimate is:

$$\hat{\theta}_a = \frac{1}{n}\sum_{i=1}^{n} \frac{g\left(X_i\right) + g\left(-X_i\right)}{2}, \qquad \text{with i.i.d. } X_i \sim N(0,1).$$

Antithetic variables will always reduce variance if the function g is monotone increasing (strictly non-decreasing).

4.2 SAMPLE RECYCLING

The sample recycling method was introduced by Feng and Li (2022). It is a method for making nested simulation more efficient. The main idea is that instead of running a full set of inner simulations for every outer scenario, the method performs the inner simulation only for a small set of reference scenarios. These simulated inner paths are then recycled for other scenarios by applying appropriate likelihood ratio weights, allowing the same samples to represent many similar states of the world. The broad algorithm for our purposes is as follows:

1. Simulate states of the world
    - N_{outer} paths
    - Each path has T time steps.
    - This results in a N by T matrix.
    - This step is the same as before.
2. For each time point t:
    - **Select B outer paths at time t (reference paths)** (Do not fix them globally.)
        - They should cover the simulation well – use quantiles.
    - **Run the inner simulations using the reference paths**
        - For each reference, run N_{inner} inner paths.
        - For each inner path, save the simulated value at $t+1$ and the future path info needed for t to T (for Asian options this is the sum of future prices on the path).
    - **Now, no new inner simulations are done. Instead, the existing ones are recycled.**
        - Assign each new outer simulation to its nearest reference.
        - Compute one *weight* per inner path

- *Rebuild the payoff* on the same inner path, but using the new outer path's own past at t.
- Use the weights to get a weighted average derivative value for the outer path at t

- These values then make up the mark-to-market matrix.

The weights when using a GBM are:

$$w_{i|r,j} = \frac{\exp\left[-\frac{(\ln(Y_j/S_i(t))-\mu)^2}{2\sigma^2\Delta t}\right]}{\exp\left[-\frac{(\ln(Y_j/S_r(t))-\mu)^2}{2\sigma^2\Delta t}\right]} = \exp\left[-\frac{(\ln(Y_j/S_i(t))-\mu)^2-(\ln(Y_j/S_r(t))-\mu)^2}{2\sigma^2\Delta t}\right],$$

where Y_j is the first inner step for inner path j $\mu = \left(r - \frac{1}{2}\sigma^2\right)\Delta t$, i is the outer simulation number and r is the reference.

4.3 COMPUTATIONAL EFFICIENCY

Another way of improving the Monte Carlo simulation is through computational efficiency. Since Monte Carlo is inherently computationally expensive, it is therefore crucial to optimise the implementation from a coding perspective (Green, 2015). An easy way to improve efficiency is to use vectorisation (array programming), where an operation can be applied to an entire vector or matrix. In Python this is easily done with NumPy or Pandas. Taking this a step further is multiprocessing. A computer CPU has multiple cores. Multiprocessing is when different applications run in parallel on different cores. This is a lot faster, as tasks get done at the same time instead of sequentially (De Prado, 2018). In the context of EPE, several nested simulations could be run at once instead of one at a time. From a Monte Carlo perspective, there is also research on the optimal simulation size of the outer and inner loops in a nested simulation (Feng and Li, 2022). Finally, there is a choice of programming language. Python is a powerful language, especially in financial modelling. However, it is not that fast. According to Green (2015), C++ is most likely the best choice from a speed perspective on CPU-based platforms.

CHAPTER 5

NUMERICAL EXAMPLES

The following examples are based on a payer interest rate swap, a European option and an Asian option. Interest rate swaps are incredibly popular and thus important. Asian options allow us to run a nested simulation, with European options being used for comparison. All simulations were done in Python with 10000 (outer) paths.

5.1 SIMULATING RISK FACTORS

5.1.1 Interest Rates

As mentioned before, we model interest rates with the HW1F. The stochastic differential equations as given in Hull (2016) are:

$$dr_t = [\theta(t) - ar_t]dt + \sigma(t)dW,$$

where a is the mean reversion parameter, dW is a Brownian motion and $\theta(t)$ is calibrated to the yield curve. The simulation was done using QuantLib, which is based on Brigo and Mercurio (2006). Here we show an example simulation, first showing the yield curve and then the simulated interest rate paths:

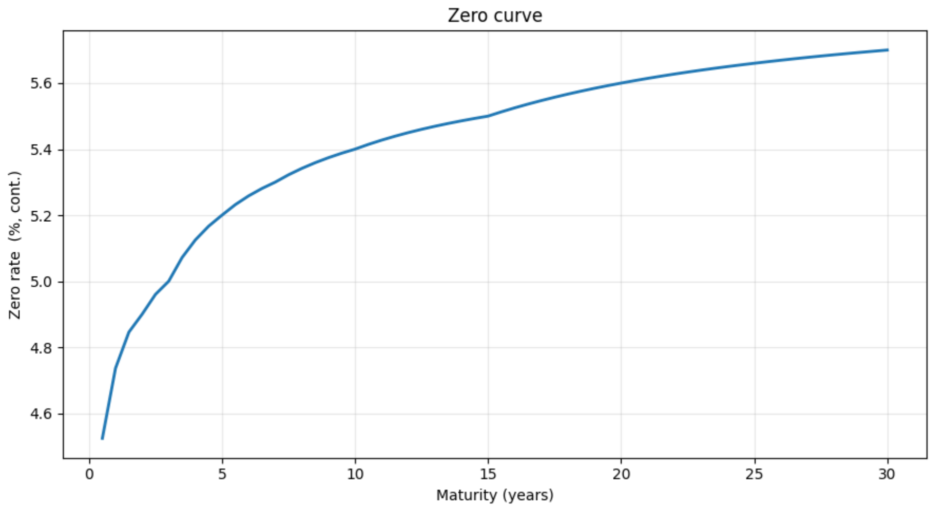

Figure 5.1: Example yield curve

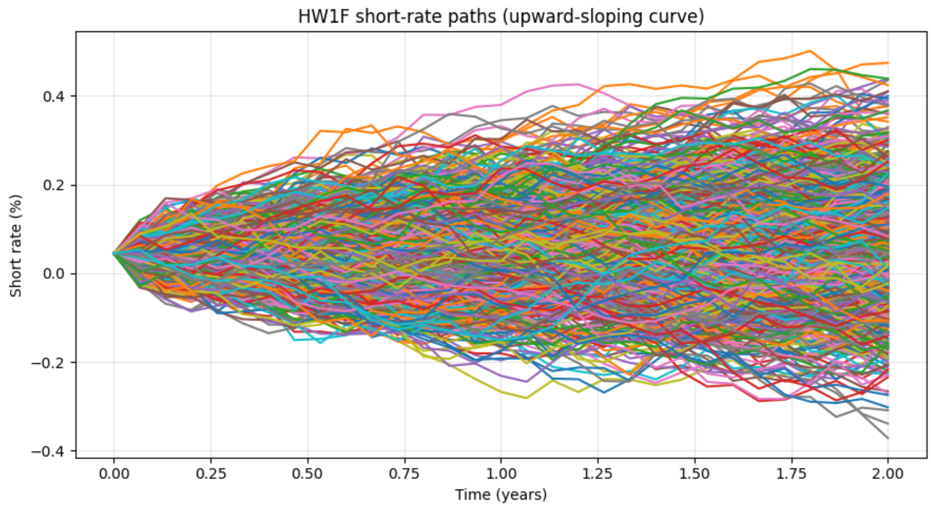

Figure 5.2: Simulated interest rates

Here you can clearly see the drawback of HW1F, where the rates go negative.

5.1.2 Equity Prices

Equity prices are simulated using a geometric Brownian motion as given in Hull (2016):

$$dS = rSdt + \sigma SdW$$

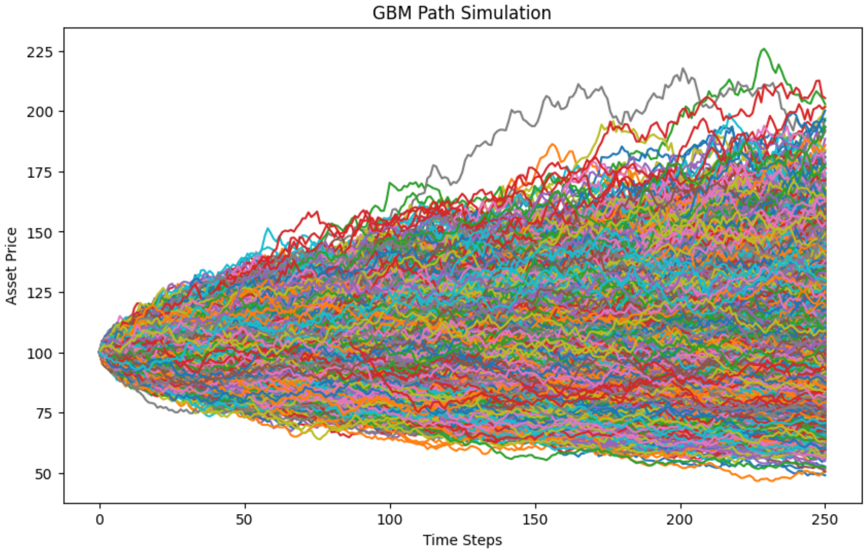

Figure 5.3: Simulated equity prices

5.2 EXPOSURE PROFILE SIMULATION

We show the EPE profile plots over time for the derivatives.

5.2.1 Interest Rate Swap

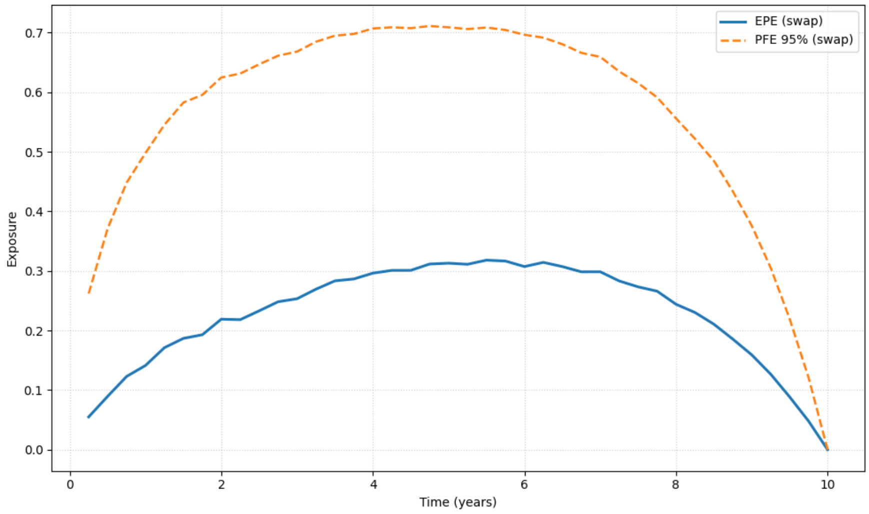

Figure 5.4: Swap EPE and PFE Profile

5.2.2 European Option

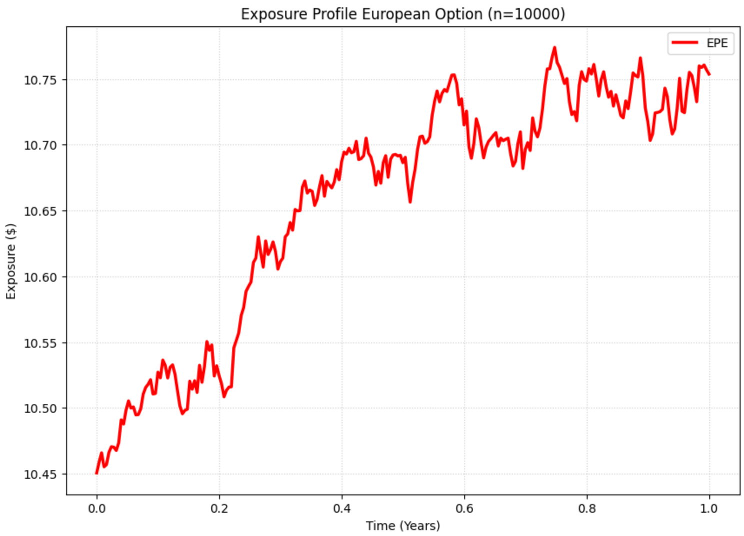

Figure 5.5: European option EPE profile

5.2.3 Asian Option

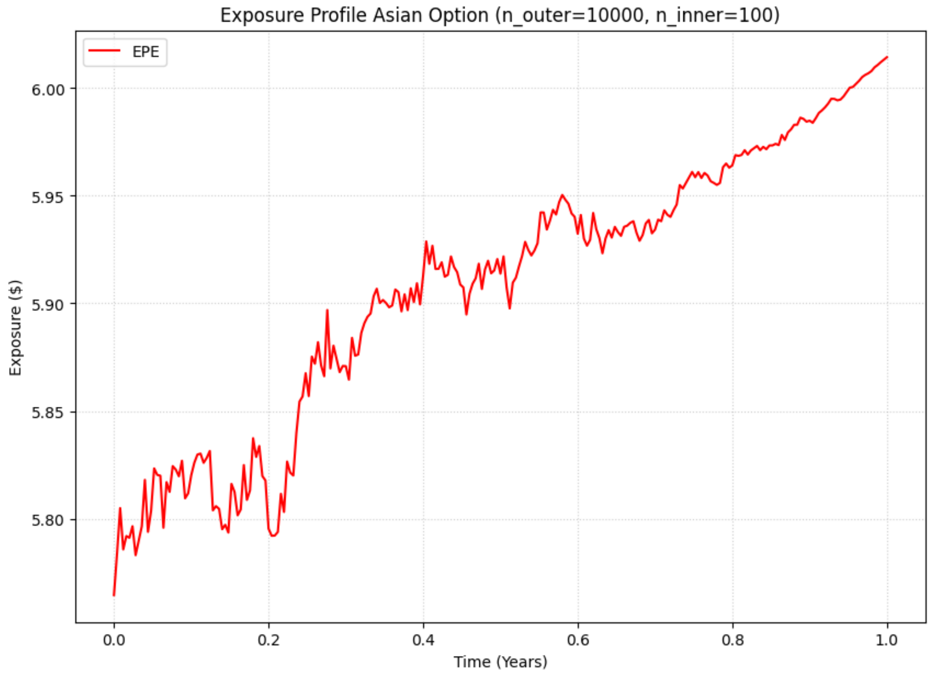

Figure 5.6: Asian option EPE

For the Asian option, each inner simulation consisted of 100 paths. It took less than a second to calculate the exposure profile for the European option and 814.0936 seconds for the Asian option. This shows how slow nested Monte Carlo can be.

5.3 VECTORISATION

The importance of vectorisation is likely obvious, as applying something to a whole row or column at once is clearly much quicker than a crude implementation. To give a basic illustration, the EPE simulation for European options was run twice, each with 10000 simulations and 250 time steps. The first one used a double for loop to simulate an equity price at each state and then a double for loop to price the option at each state. This took 125.9185 seconds. The second was vectorised and took only 0.2146 seconds.

5.4 ANTITHETIC VARIABLES

For the antithetic variables, we start by looking at the exposure profiles of the crude versus the antithetic methods.

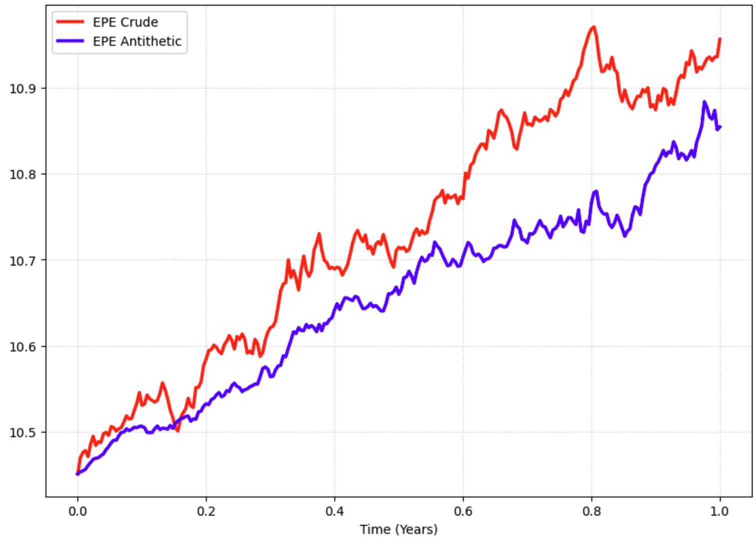

Figure 5.7: European Crude v.s. Antithetic EPE

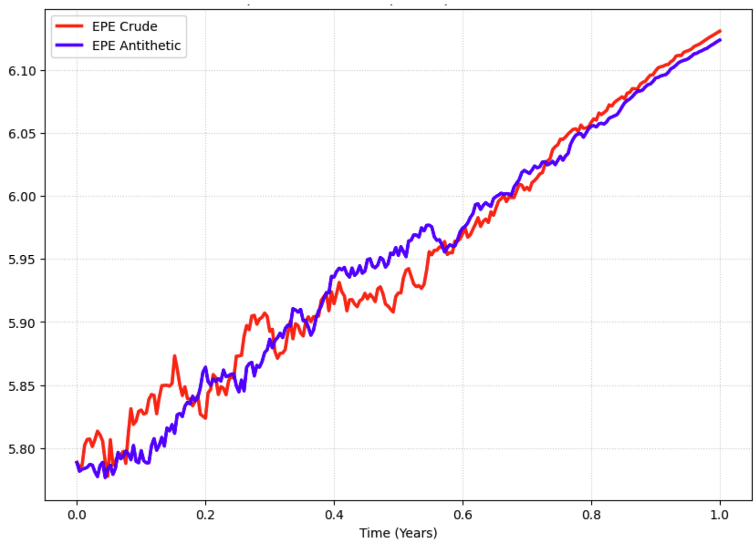

Figure 5.8: Asian Crude v.s. Antithetic EPE

We also look at the standard errors over time:

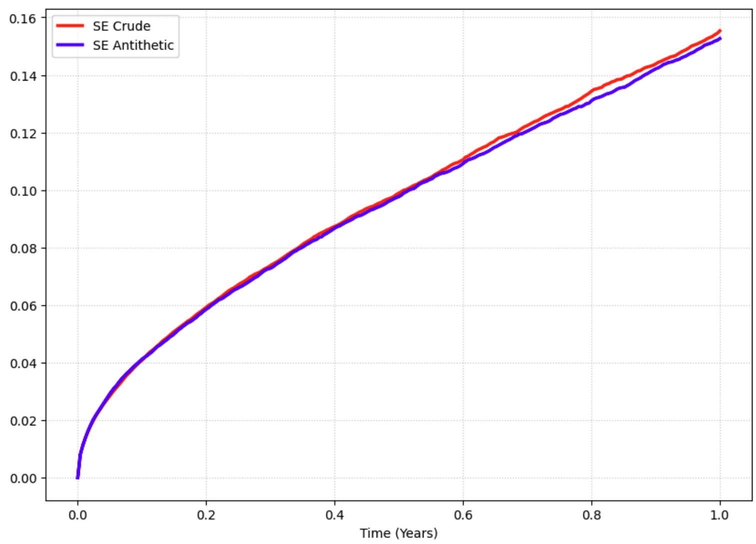

Figure 5.9: European Crude v.s. Antithetic SE

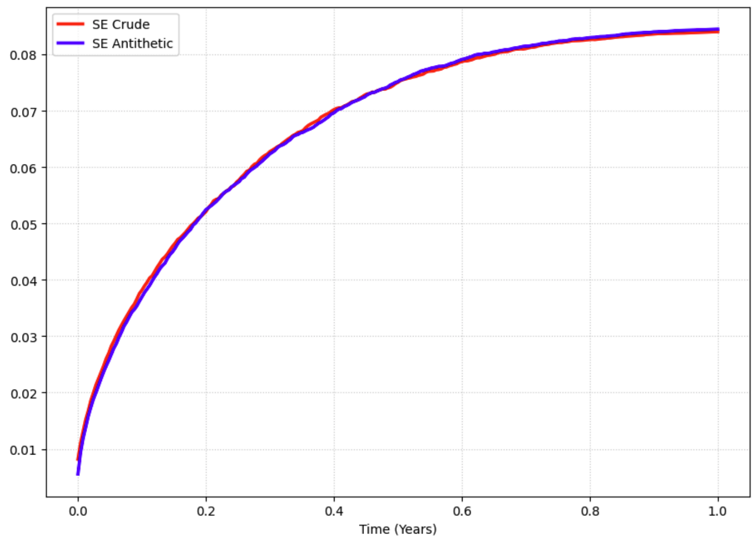

Figure 5.10: Asian Crude v.s. Antithetic SE

We can clearly see from the plots that antithetic variables didn't make any real difference from an accuracy perspective. Additionally, the antithetic approach was only about 5% faster.

5.5 SAMPLE RECYCLING

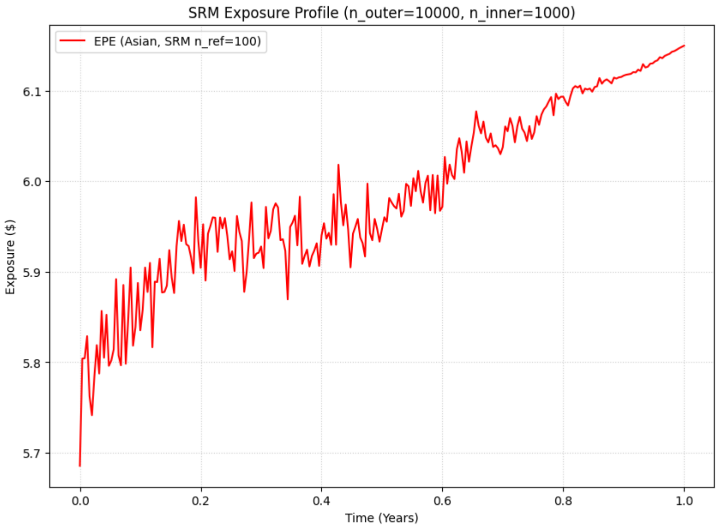

Figure 5.11: Asian Option SRM EPE

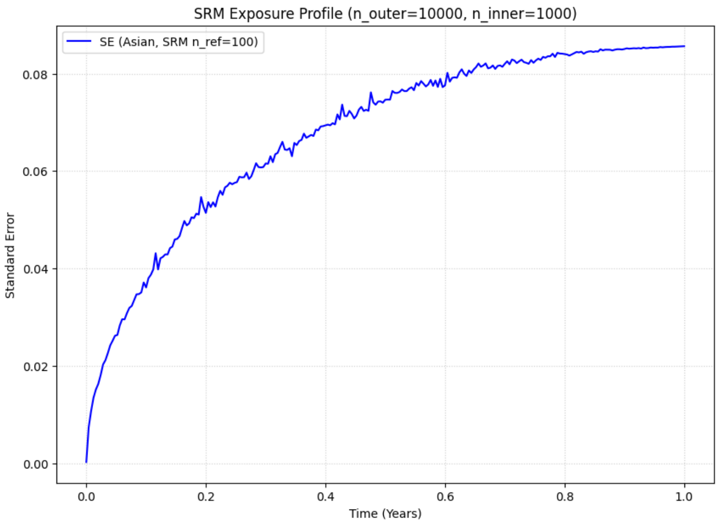

Figure 5.12: Asian Option SRM SE

Here 100 reference paths were used, and each inner simulation has 1000 paths. We see the EPE and standard error profile for the sample recycling methods. EPE heads in the same direction as the crude method. The standard error is a little bit jagged; however, the magnitudes seen seem to be equivalent to the crude method, meaning that the sample recycling method is still accurate. The big upside to this is that it only took 166.8959 seconds to calculate, compared to the 814.0936 seconds for the crude method – almost five times faster!

5.6 MULTIPROCESSING

Lastly, we compare the crude nested Monte Carlo for the Asian option's EPE with a parallelised implementation. As we know, the crude method took 814.0936 seconds. The parallelised method took 178.62 seconds.

CHAPTER 6

CONCLUSION

This project explored the simulation of future exposure, the computational challenges that come with it, and several practical methods to address these challenges. Through theoretical discussion and numerical examples, it is clear that traditional Monte Carlo simulation, while flexible and robust, becomes highly computationally expensive as the dimensionality of the problem increases. The results demonstrated that by implementing more efficient simulation techniques—such as vectorisation, multiprocessing, and advanced methods like the sample recycling algorithm—it is possible to achieve significant reductions in runtime without sacrificing accuracy. These improvements not only make nested simulations more feasible for complex derivatives but also highlight the importance of efficient implementation in exposure modelling. Overall, the findings show that careful algorithmic design and parallel computation can substantially accelerate exposure simulations, making real-time or near real-time xVA estimation far more attainable in practice.

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